Robotics and Flexible Automation

Prof. Zengxi (Stephen) Pan

BJTU 2026



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Industry Cases (not project tasks)

Facility for Intelligent Fabrication



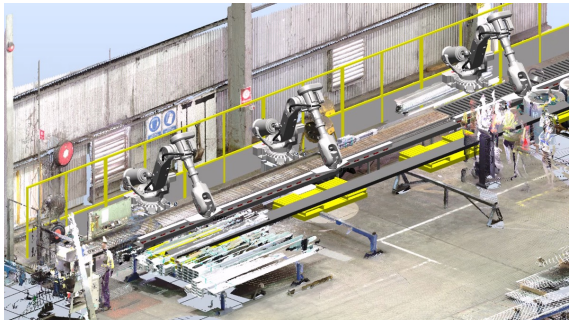
VR / AR



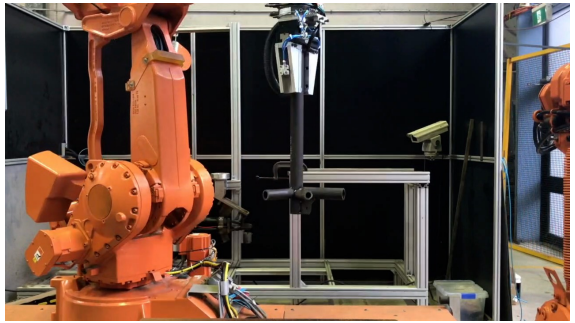
Machine Sensing



Materials Science



Robotic Automation



Welding / Fabrication

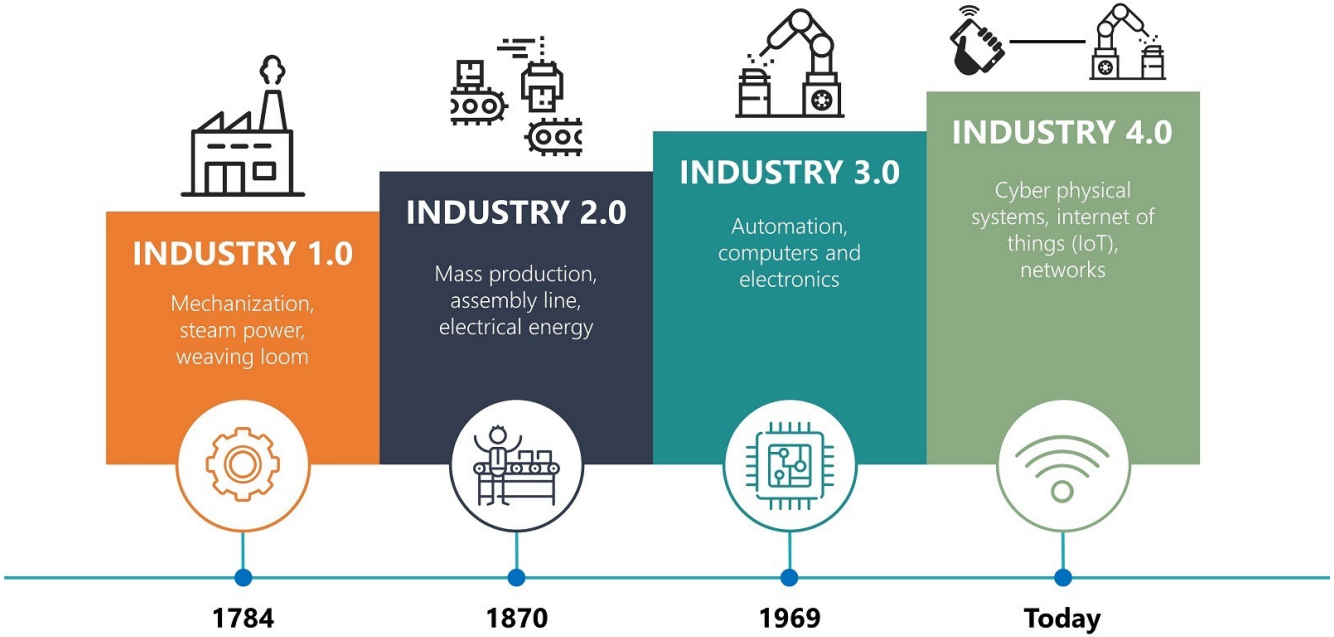


Material Handling

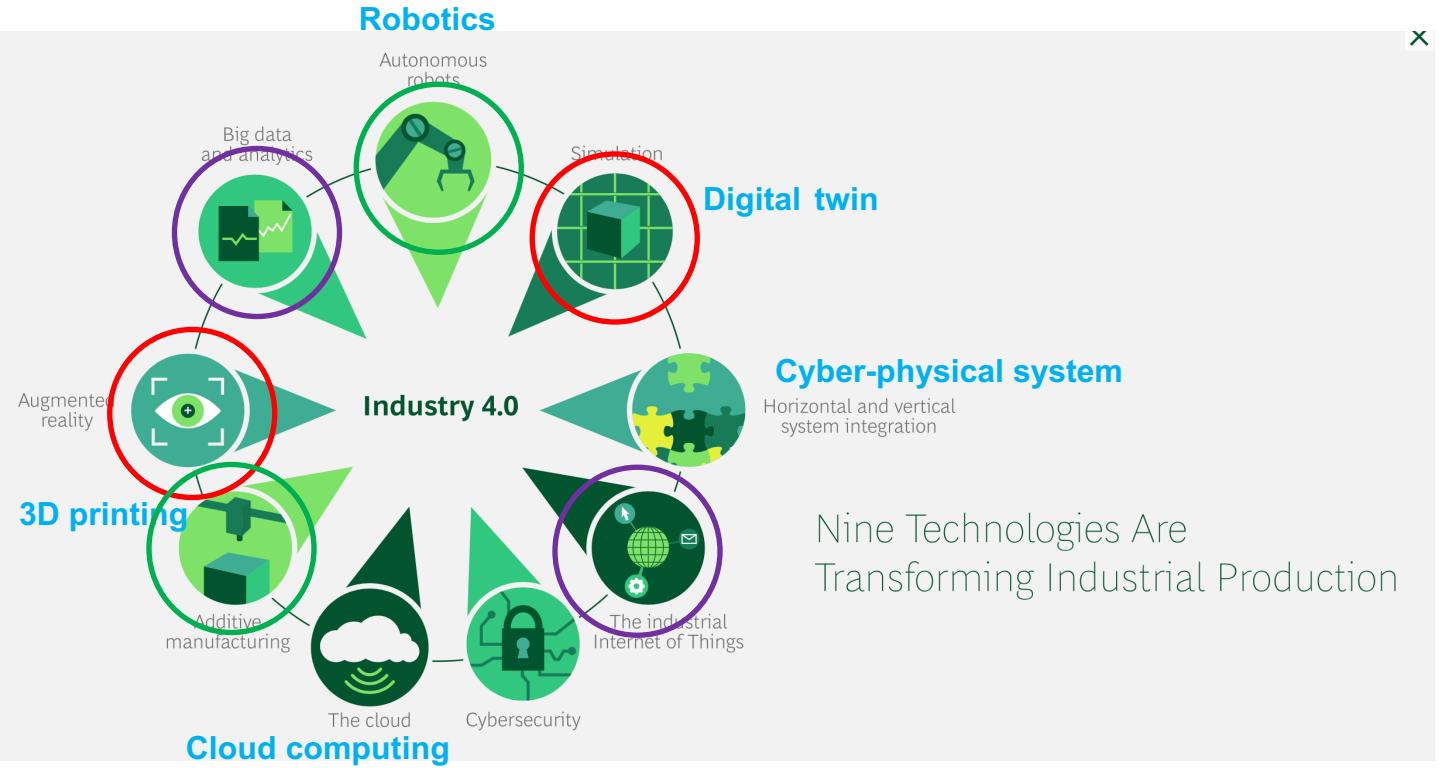
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Industry 4.0

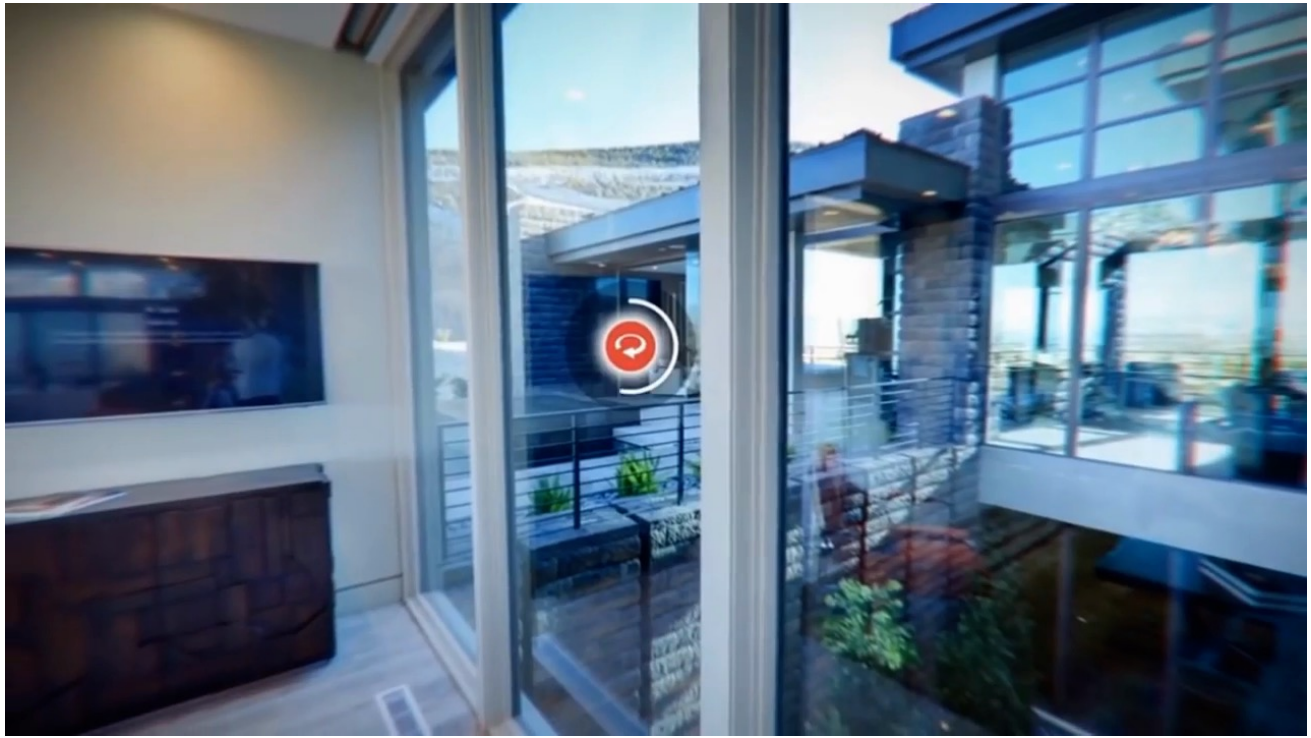
From 3.0 to 4.0



9 Pillars of Industry 4.0



9 Pillars of Industry 4.0



Tesla Manufacturing Plant

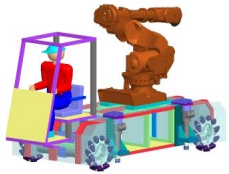


Lean Automation

What is Lean Automation?



1. Flexible manipulation and light weight robotics



6. Re-locatable mobile platforms and workstations

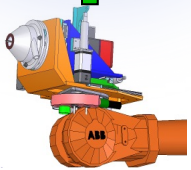


2. Offline programming and digital integration

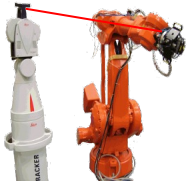
High Accuracy Lean Automation



5. Flexible or fixtureless tooling



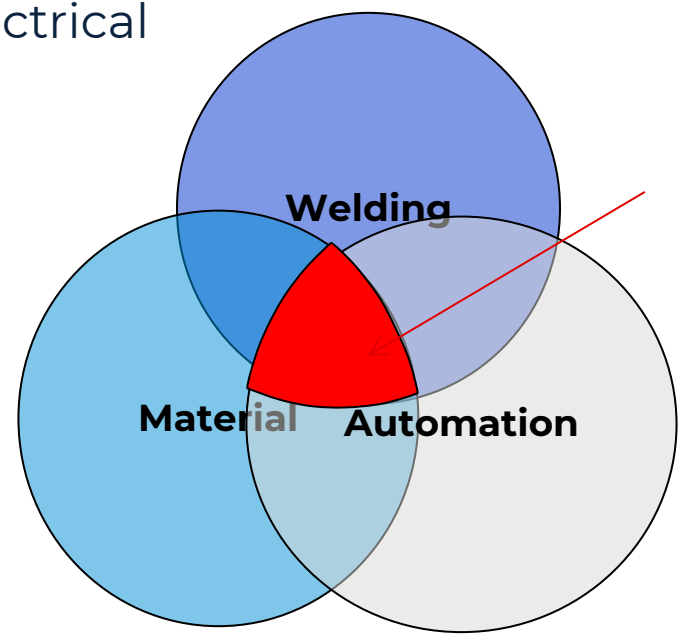
4. Purpose built intelligent and multifunctional end effectors



3. Metrology Feedback and calibration tools

Research Group

- Applied Automation / Welding Engineering
- Involved in CRCs, Defence Materials Technology Centre (DMTC)
- Interdisciplinary – metallurgy, automation, mechatronics, control, electrical



AOLP
WAAM

- ✓ Welding Research (1992~2012)
- ✓ Wire Arc Additive Manufacturing (2011~)
- ✓ Offline Programming (2009~2017)
- ✓ Industry 4.0 (2018~)

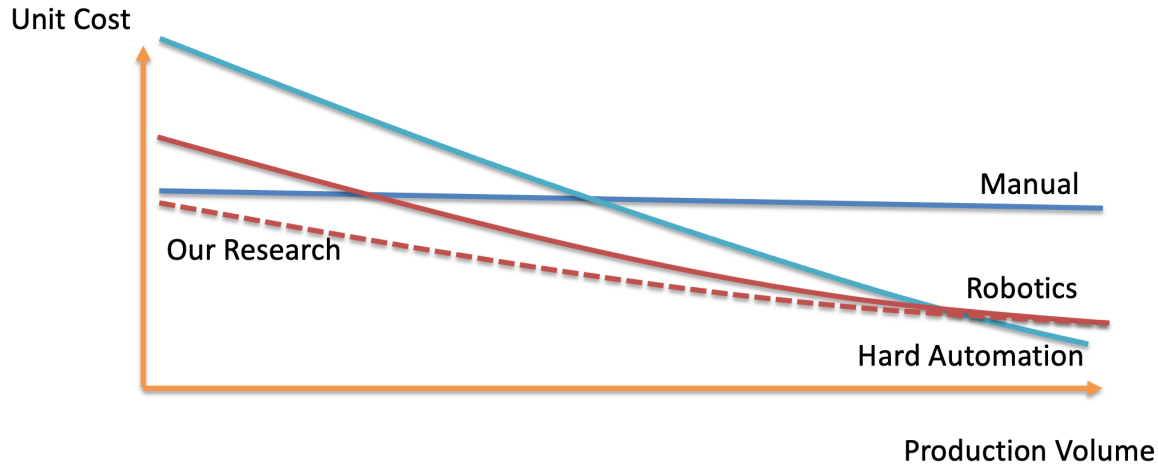
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Industrial Robotics Research

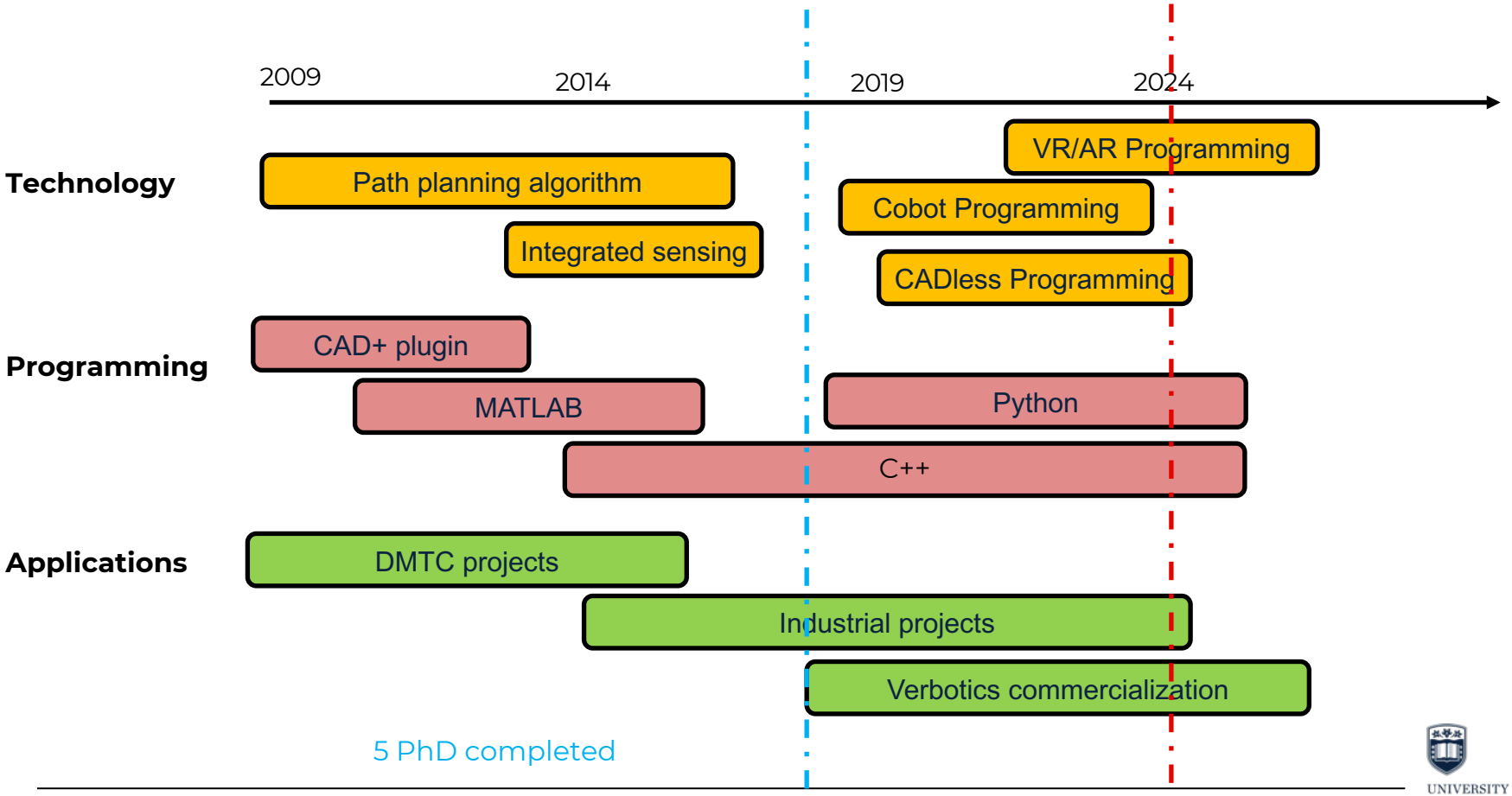
Vision

Develop robotic automation technology for SMEs

- Focus on fabrication
- Small batch volumes
- High product variation

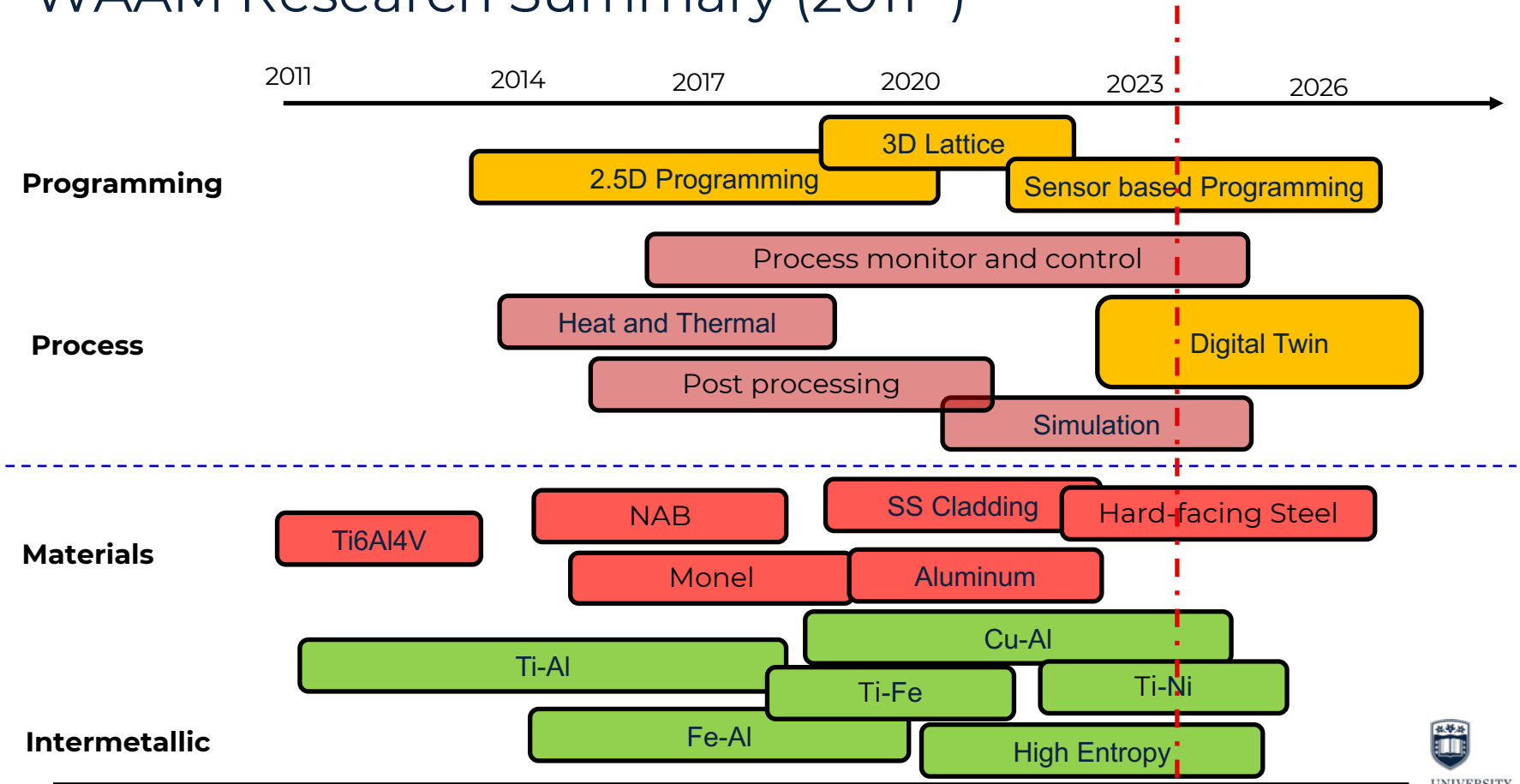


Automated Offline Programming (2009~)



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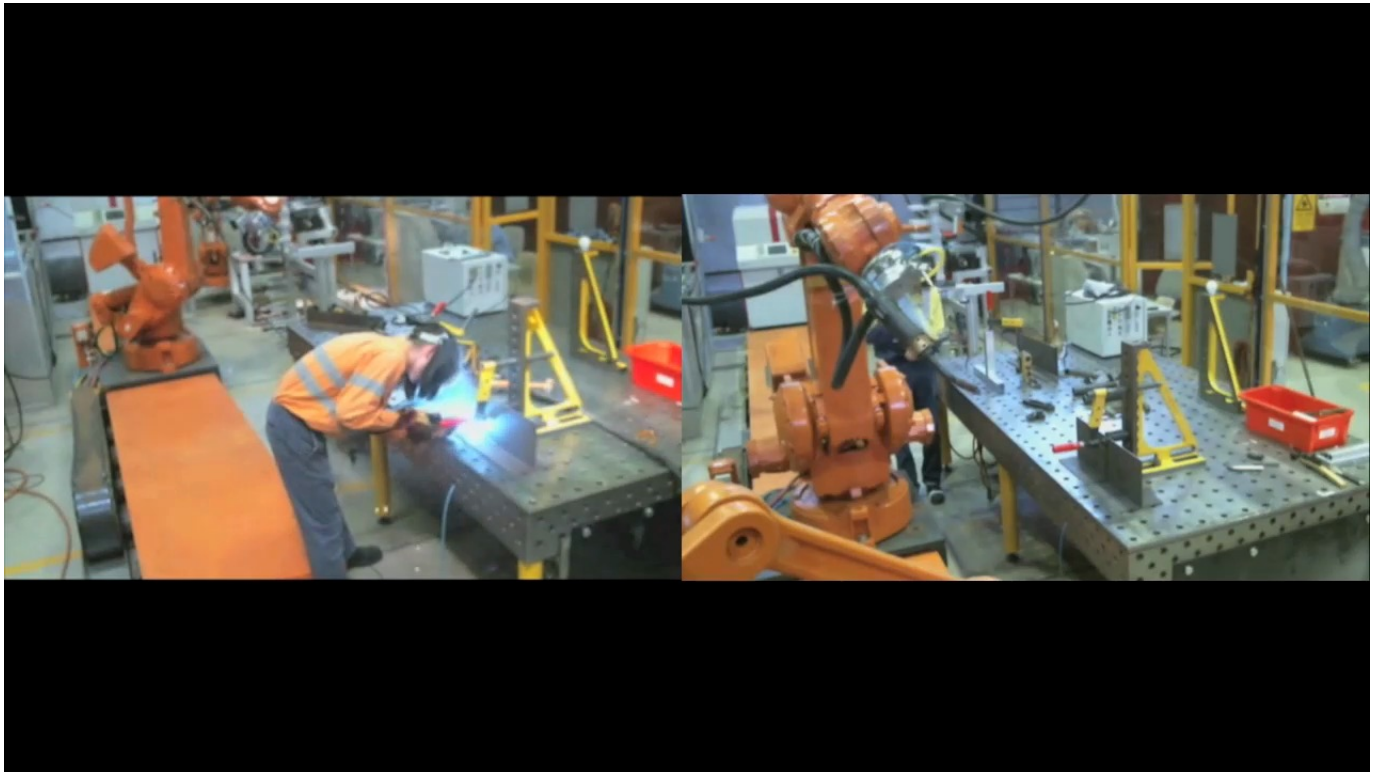
WAAM Research Summary (2011~)



14 PhD completed, 7 on going

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Video Robot Programming



Offline Programming

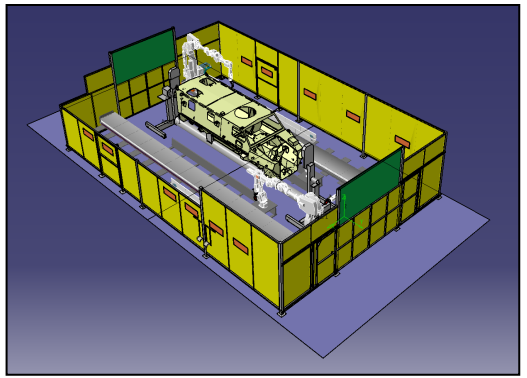
Advantages of OLP:

Robots can be programmed without taking the work cell out of production

Simulation can check for robot collisions and other unforeseen events.

It is easy to modify the program for small changes.

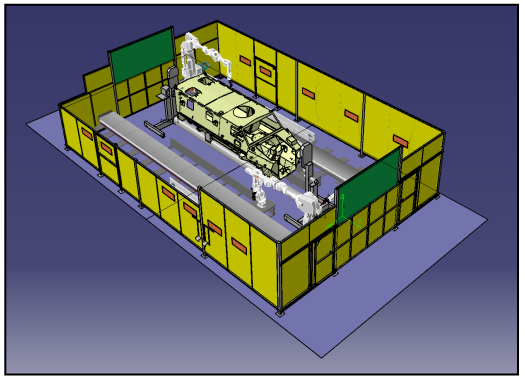
OLP and simulation can be completed with a product design such that manufacturing issues can be identified at an early stage.



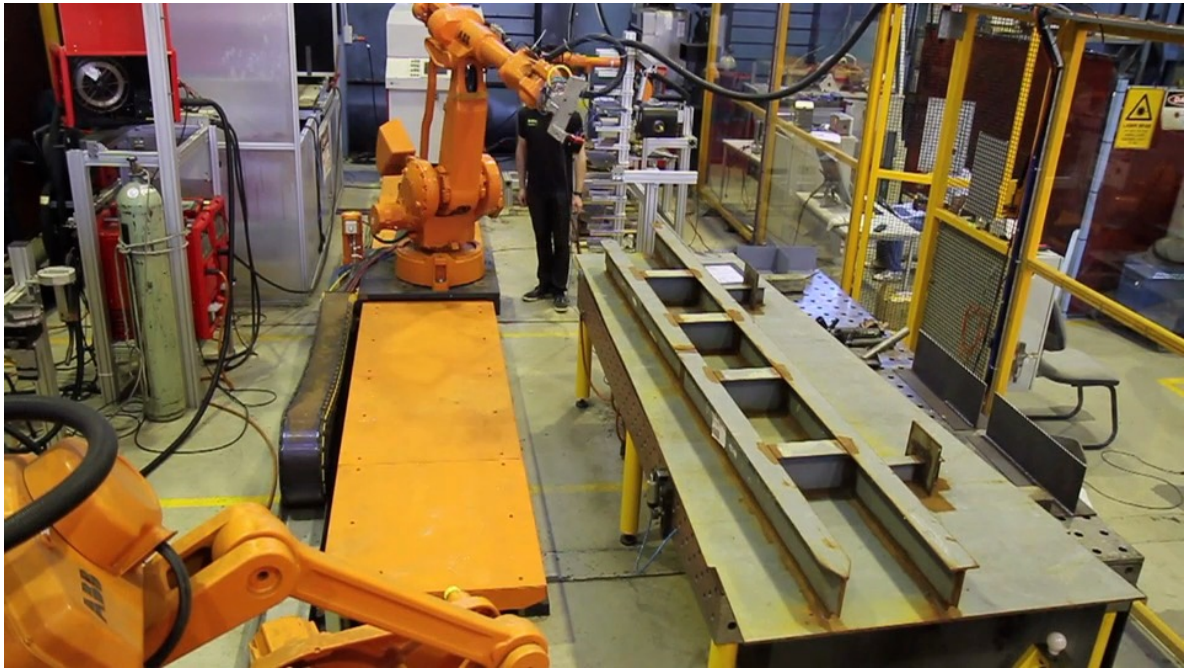
Offline Programming

Challenges of OLP:

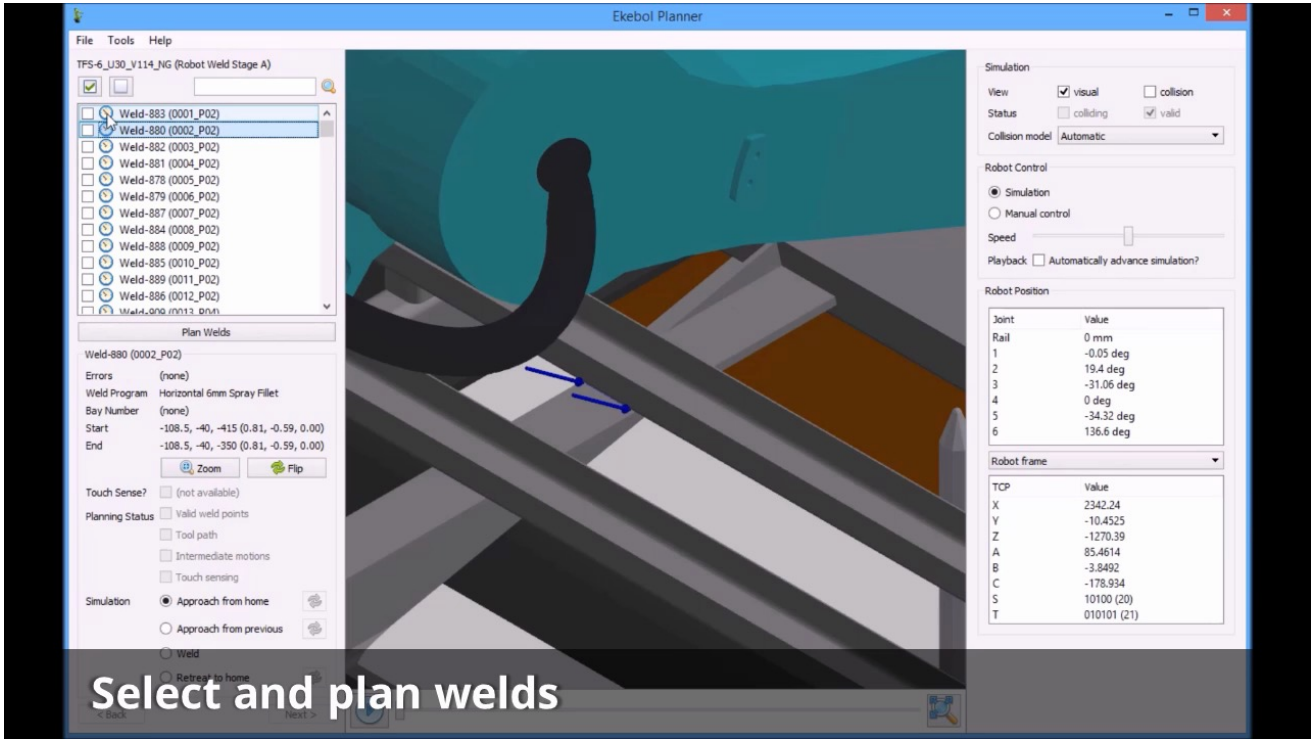
- Calibration is required to match nominal CAD dimensions with actual work cell.
- Trajectory Planning with path optimisation is non trivial and is a research area requiring add-on software tools.
- Collision avoidance also requires add-on software tools.



Sensor based Robot Programming

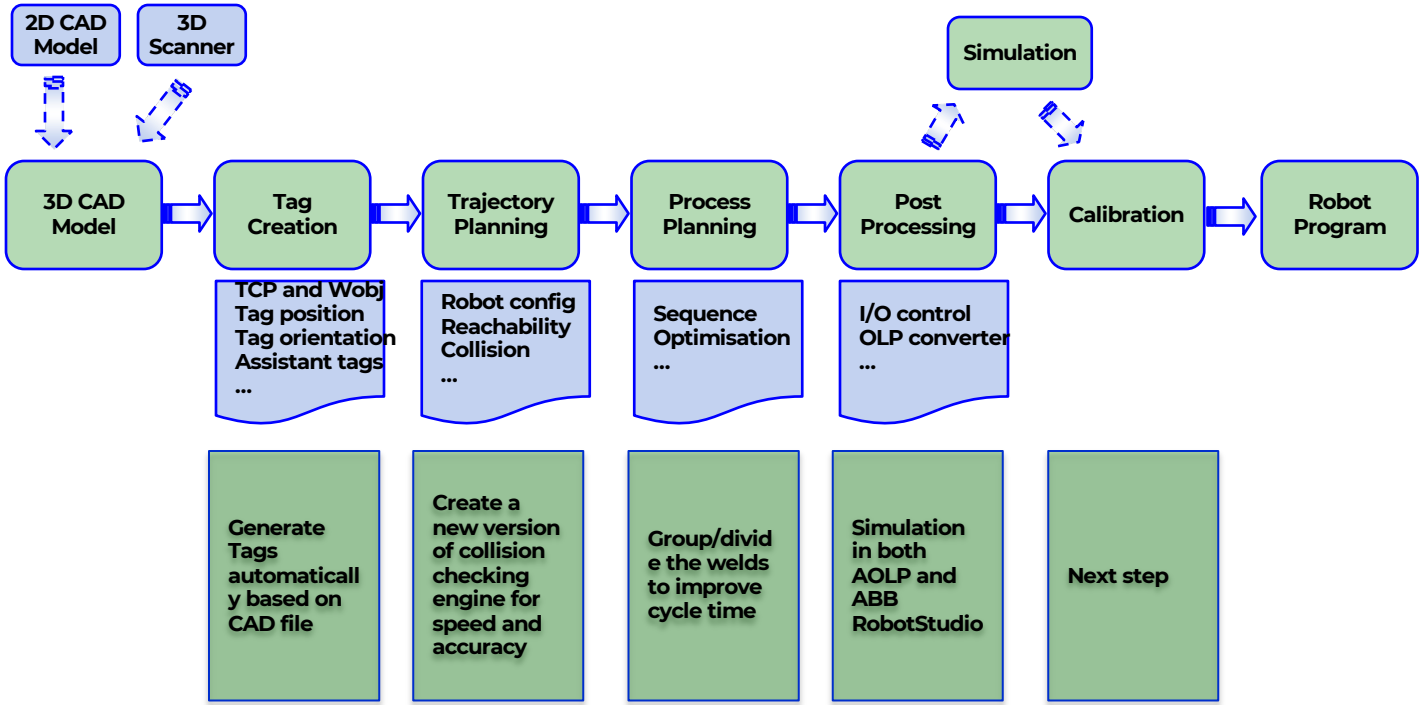


https://www.youtube.com/watch?v=U_EU7o1G1nc



https://www.youtube.com/watch?v=ul47KfI03R8

Software Flow Chart -- AOLP for Welding



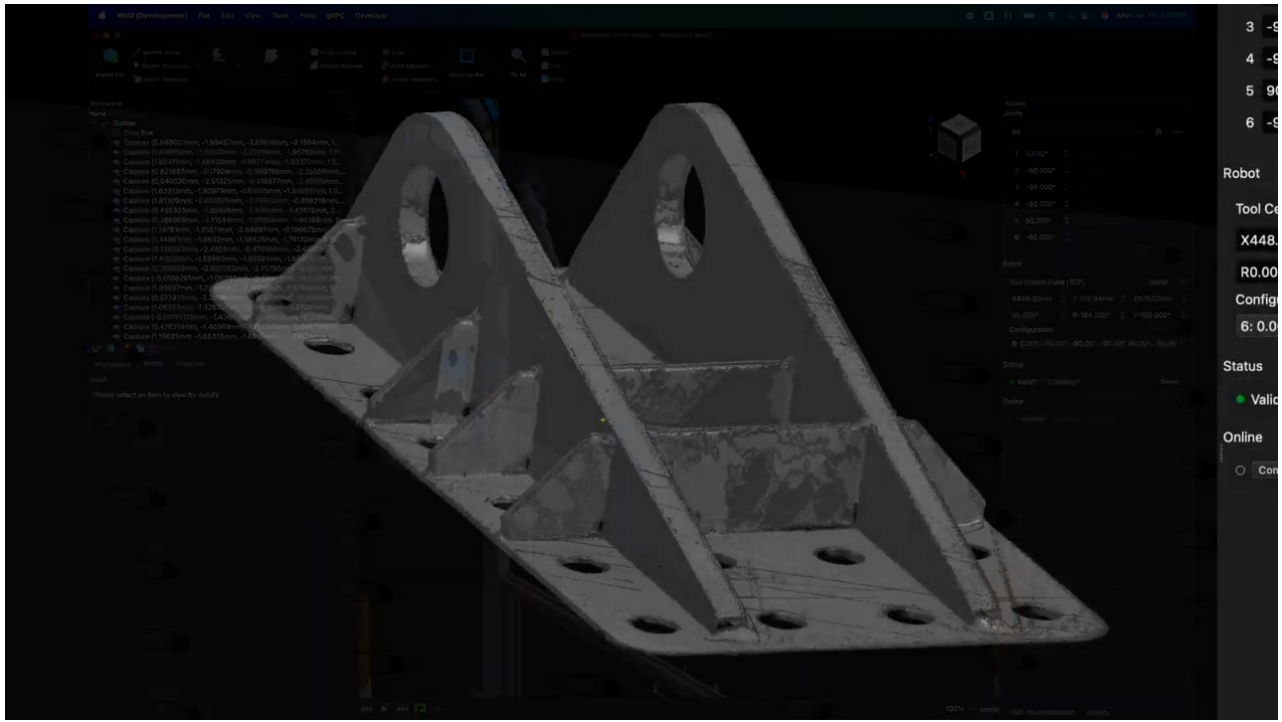
Automated Offline Programming



Verbotics Demo Videos (Arronet Boats)



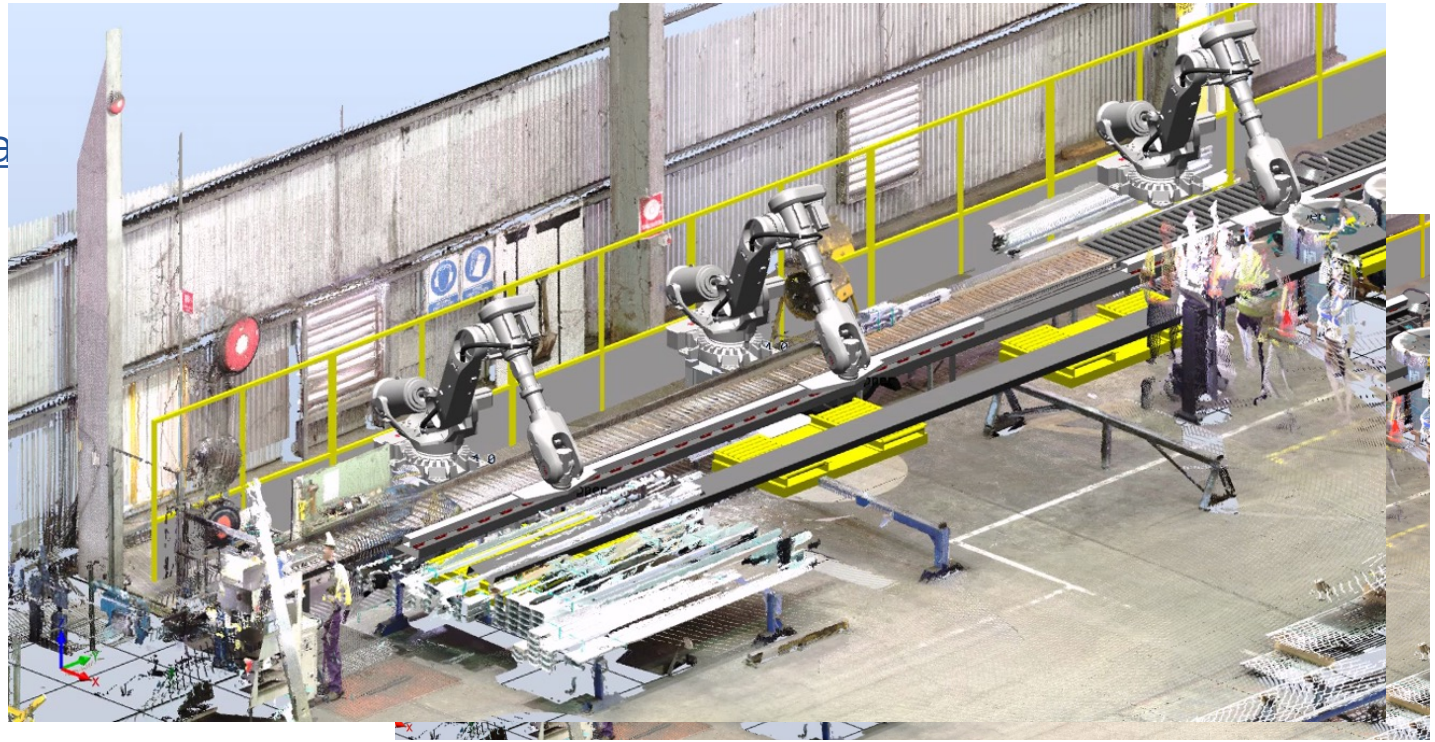
Scan to Weld



3D scan overlay with CAD

Bluescope

Beam Sta



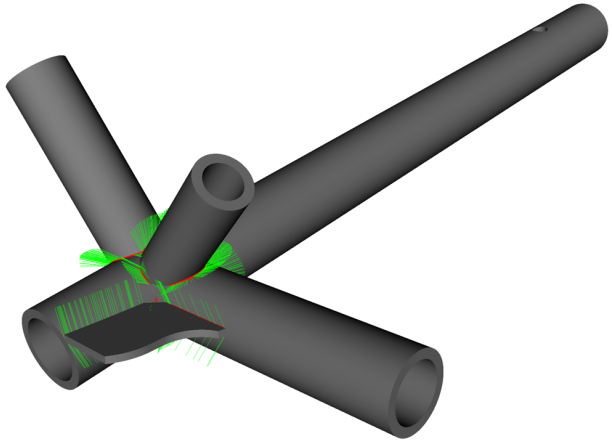
https://www.youtube.com/watch?v=aFzrwDTSCAc

Example

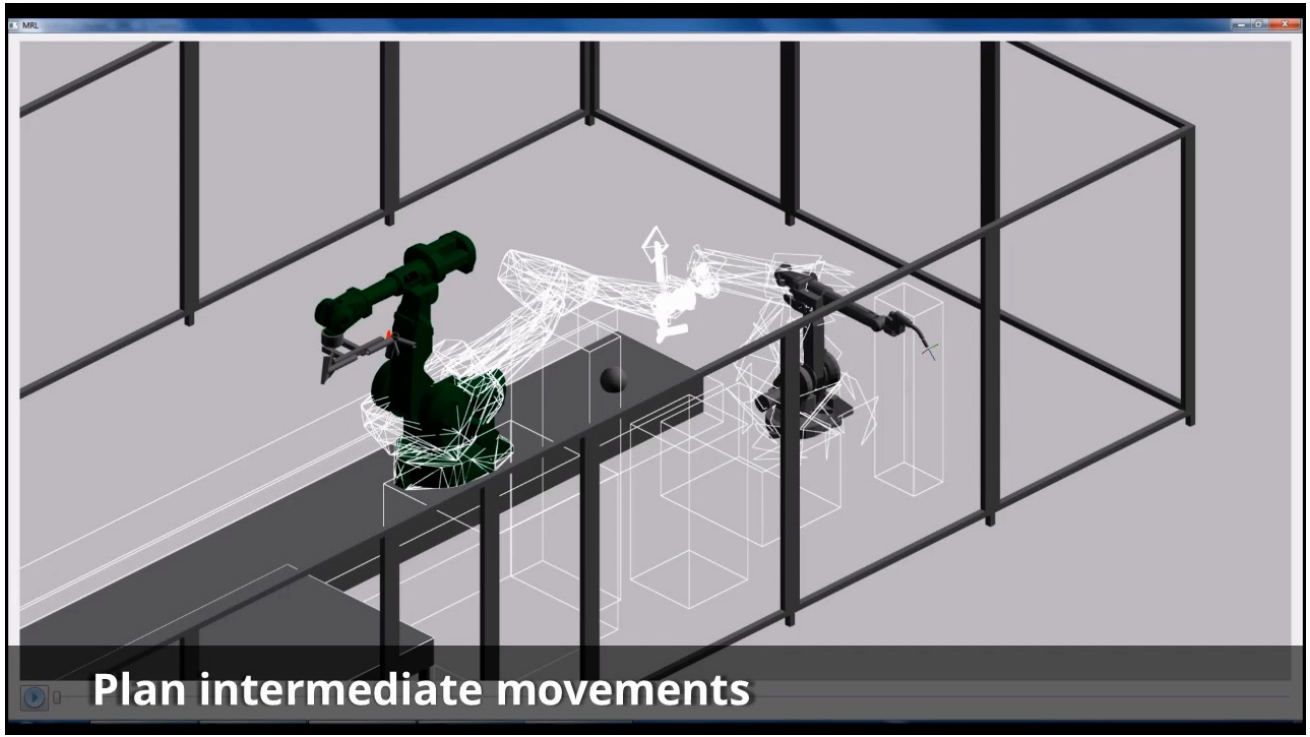
Pipe welding

UOW PipeWelding short.mp4

UOW Pipe Welding.mp4

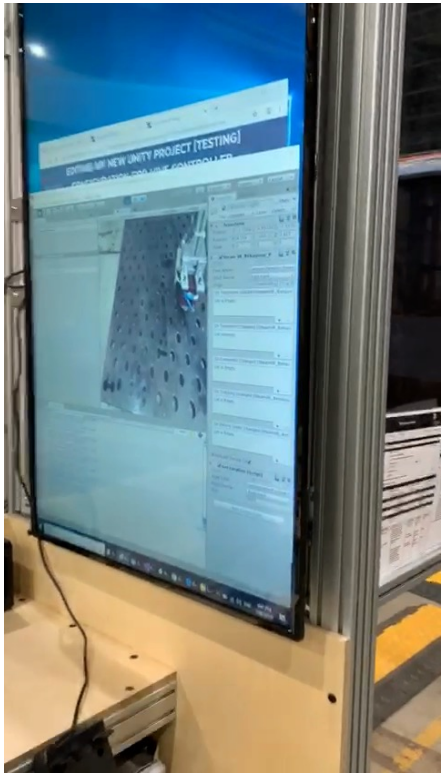




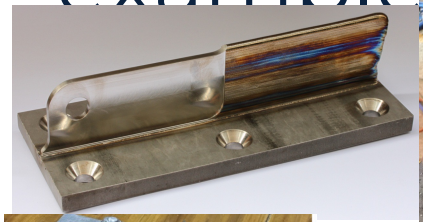


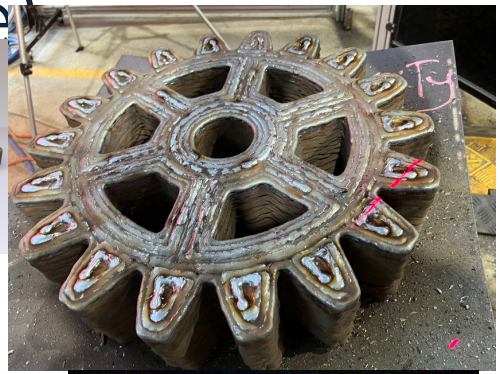
https://www.youtube.com/watch?v=F-7RCKJ9Ago

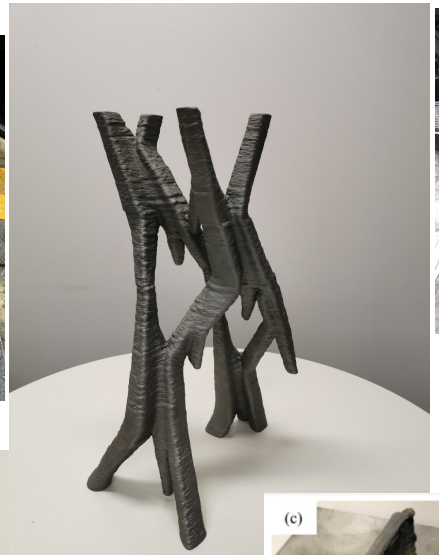
AR/VR Robot Programming



WAAM Samples (optional example)

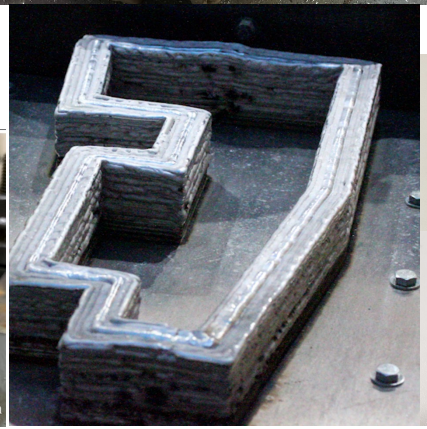


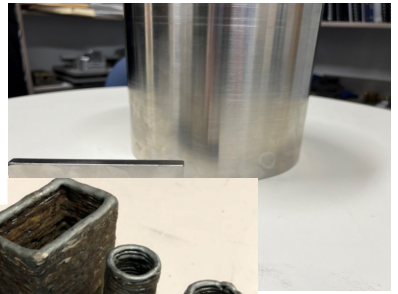


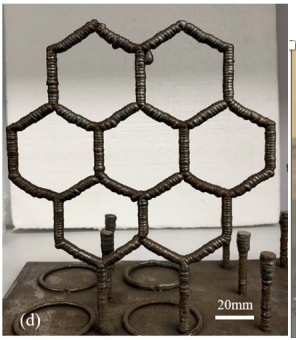


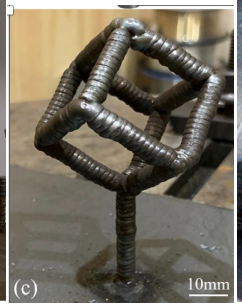






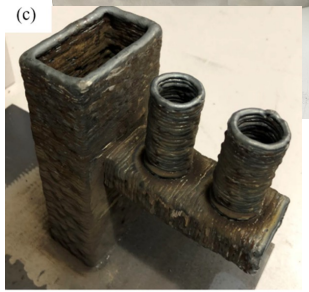










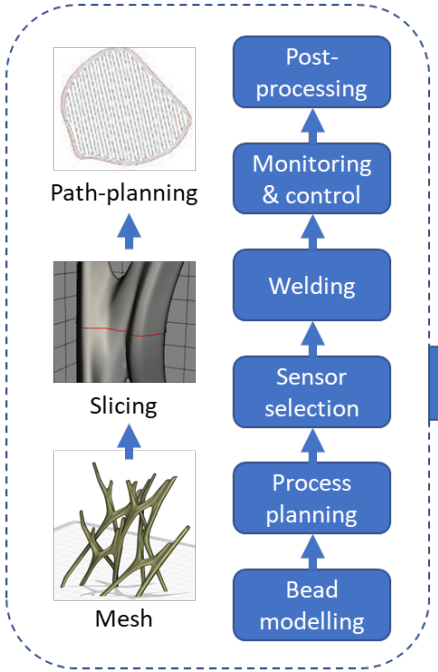


WAAM Background

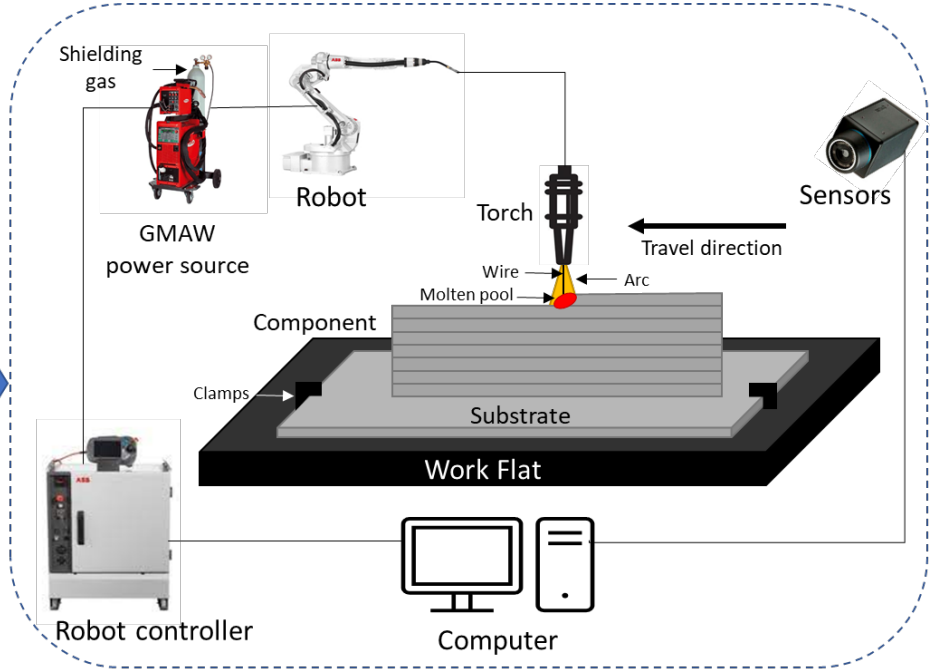
WAAM is a complicated long process time fabrication process







WAAM workflow



WAAM system

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Wire Arc Additive Manufacturing

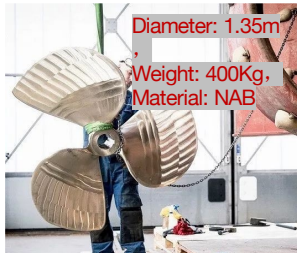
Research history and industrial progress



Complex geometry in Cranfield

Cold rolling of WAM in Cranfield





WAAM Propeller in RAMLAB

WAAM fabricate entire rocket including fuel tank and rocket engine



1990s

2013

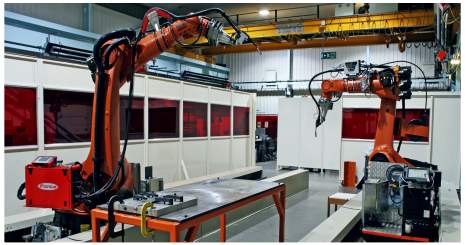
2015

2017

2021

Shape Metal Deposition by Prof. John Norrish





WAAM Software in UOW

Titanium structural components in Boeing

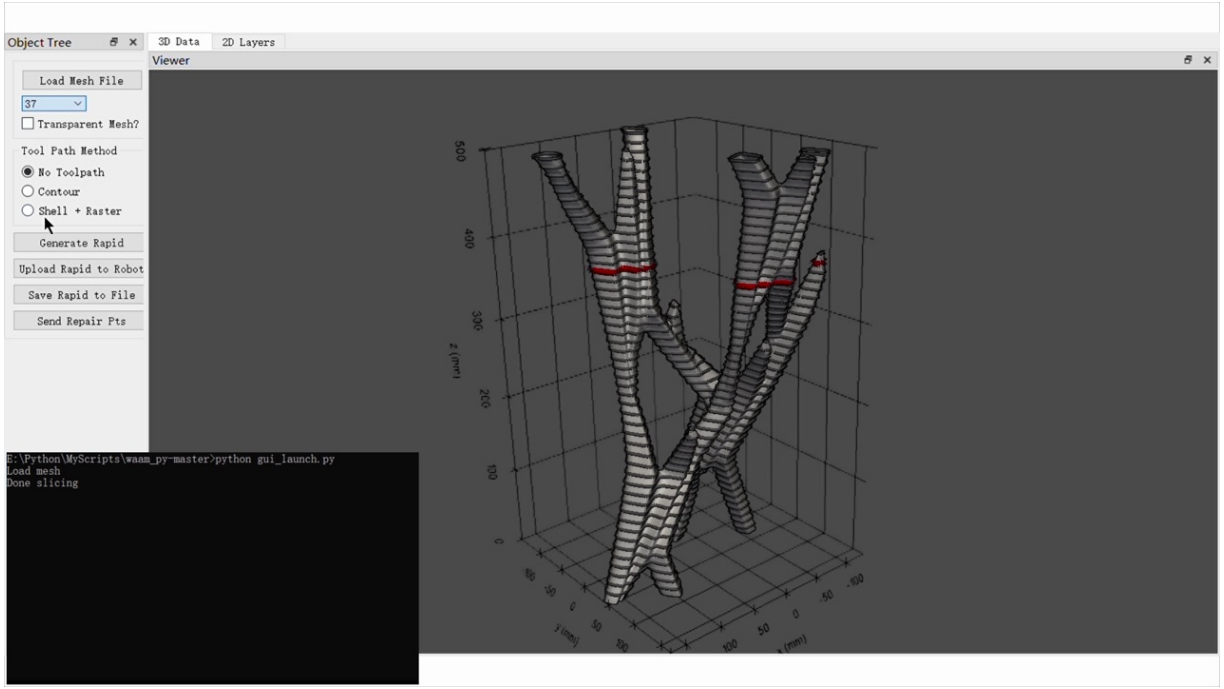


Titanium pressure vessel for space exploration



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Wire Arc Additive Manufacturing



Major Challenges in WAAM

Quality assurance

Main Challenges		
Automation System	Manufacturing Accuracy	Quality Assurance
• Path planning • Process control • Process monitoring • Automatic cooling •	• Error build-up/mitigation • Thermal management • Process parameters •	• Instability • Accuracy • Contamination • Surface profile • .....







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Current Research at UOW

Monitoring system

Signal sources of the monitoring systems

- Thermal camera: Obtain the thermal field of the deposition process.
- Voltage and current sensor: Record welding electric signals.
- Laser scanner: Obtain the 3D mesh of the deposition.
- CCD camera: Capture the image of deposited parts.
- Molten pool camera: Record molten pool behavior.



Thermal camera



Voltage and current sensor



Laser



CCD



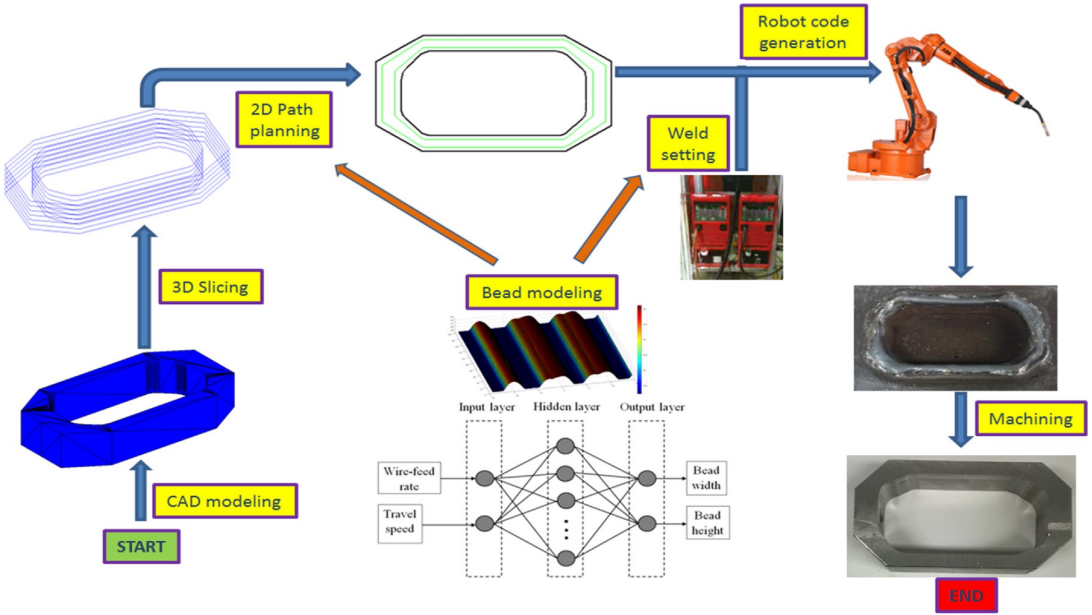
Molten pool camera

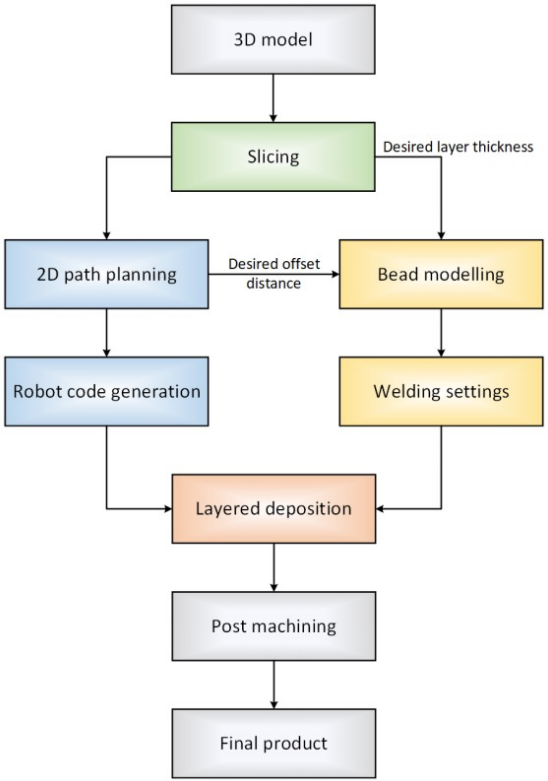
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Automatic bead modelling system

Current WAAM process

The conventional WAAM process consists of several essential steps including bead modelling, 3D slicing, 2D path planning, welding process setups, robot code generation, and post-machining process





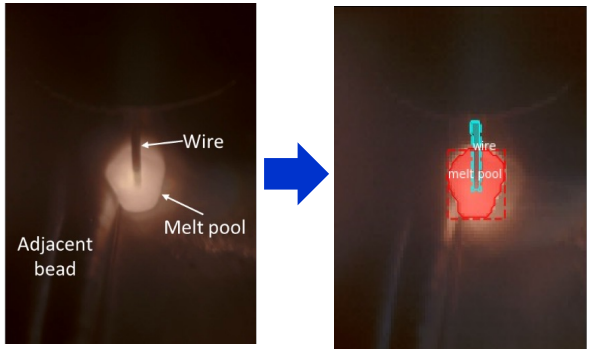
The steps of the WAAM process.

Current Research at UOW

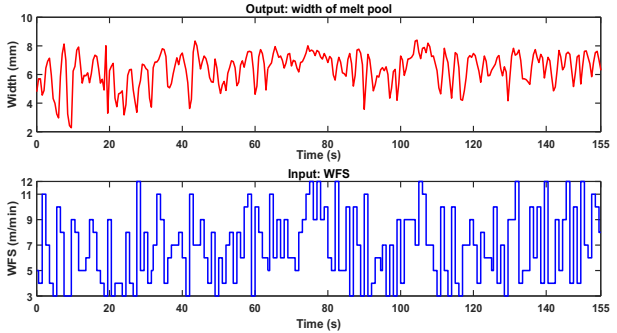
Melt pool segment and measurement using molten pool camera

Motivations:

1. **Molten pool behavior can directly reflect the bead formation.**
2. **Molten pool images are more stable to be acquired.**
3. **It is a reliable signal source in real-time monitoring system.**



Mask R-CNN based melt pool segment and measurement

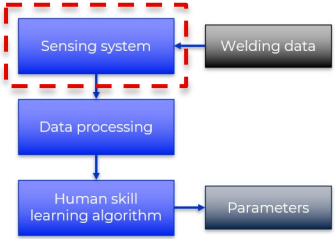


System identification experiment

Controller design…

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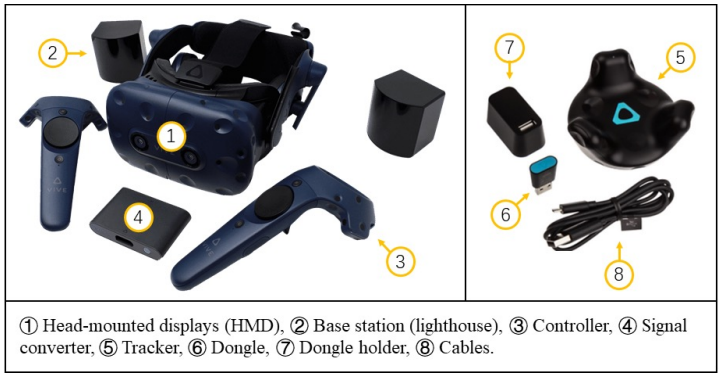
Parameter Optimization Method



Sensing system and welding parameters

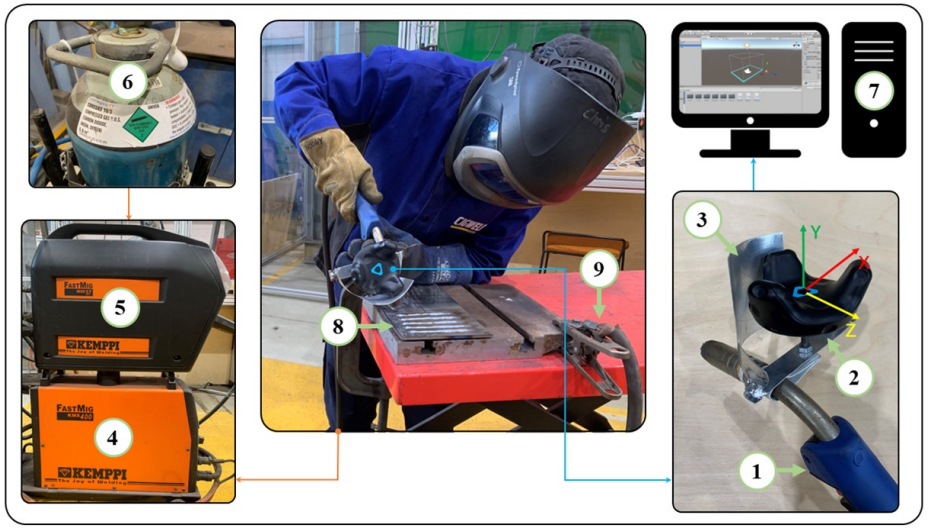
- Wire feed speed (WFS)
- Travel speed of torch (TS)
- Torch angles: work angle (WA) & travel angle (TA)

Human motion parameters



① Head-mounted displays (HMD), ② Base station (lighthouse), ③ Controller, ④ Signal converter, ⑤ Tracker, ⑥ Dongle, ⑦ Dongle holder, ⑧ Cables.

The HTC Vive VR system.



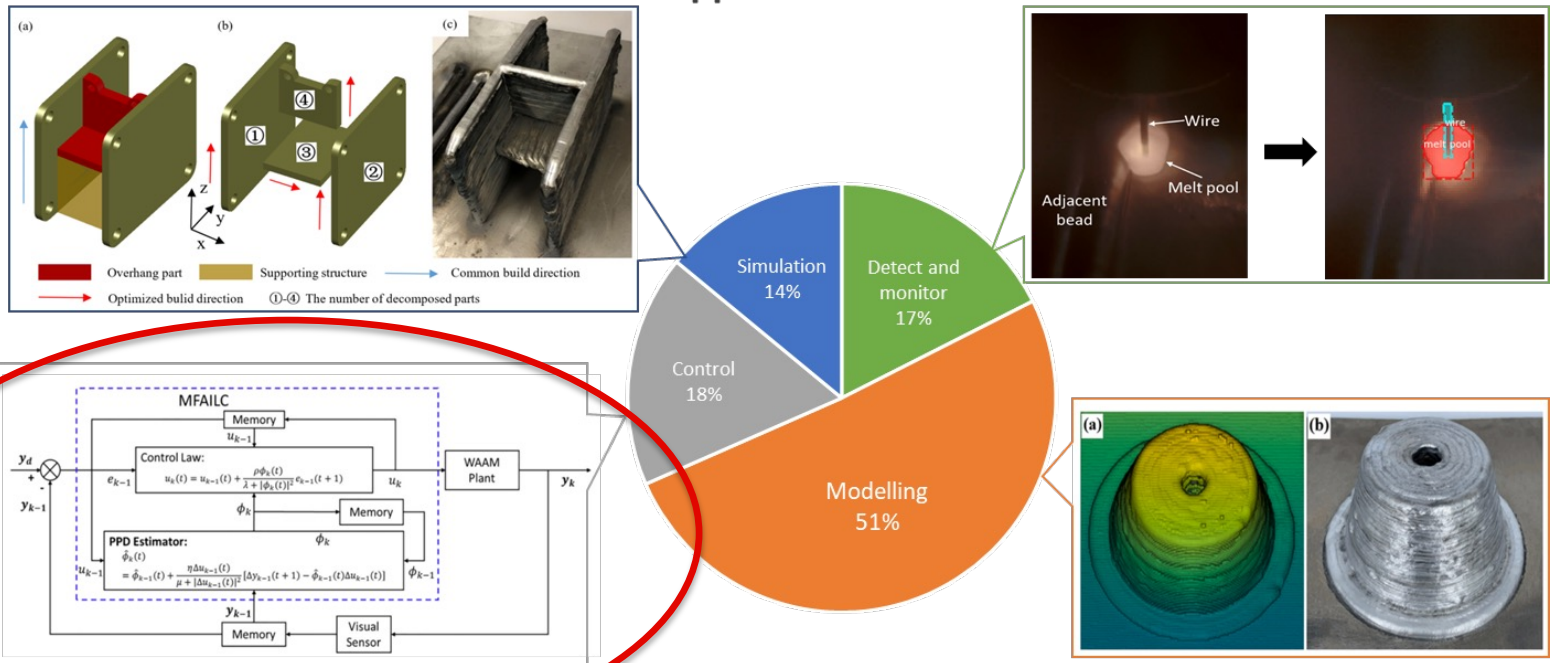
① GMAW welding torch, ② HTC Vive tracker, ③ steel shield, ④ Kemppi welding machine, ⑤ CMT wire feeder, ⑥ shielding gas, ⑦ control computer, ⑧ substrate, ⑨ protective earth wire.

The hardware setup of the human motion sensing system.

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Machine Learning in WAAM

WAAM is a complicated long process time fabrication process



ML applications in WAAM (Data source: Scopus search)

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Quality assurance

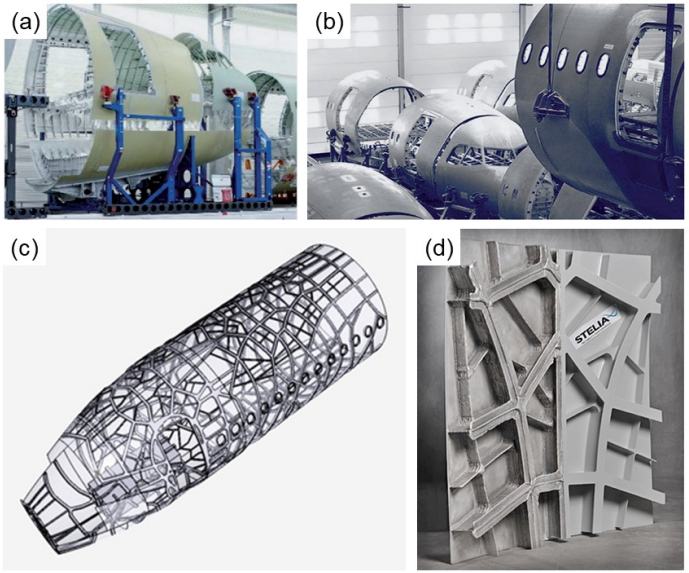
A molten pool image-based monitoring system

Challenge

To ensure the quality and reliability of high-end metal parts, monitoring the entire manufacturing process of the parts is necessary. However, it is challenging for the current monitoring methods to identify internal defects including void, porosity, and lack of fusion due to their hidden and variable nature, typically residing within the interior of parts.

A monitoring system

- Effectively identify internal defects
- Capable of detecting wide range of common WAAM defects
- Cheap, and easy to implement.

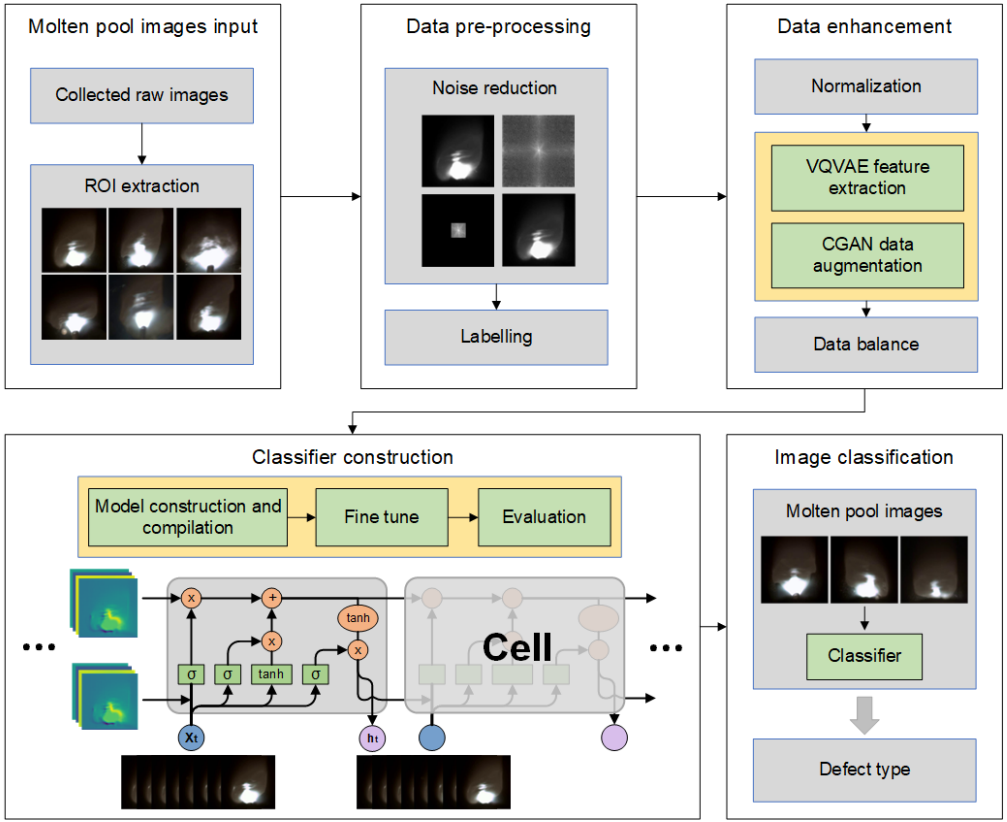


(a), (b) and (c): The aluminium airframe designs from STELIA Aerospace; (d) the close-up example of a section with complex structures manufactured using WAAM [8].

Quality assurance

System modules

- Molten pool image input
- Data pre-processing
- Data enhancement
- Classifier construction
- Image classification



The workflow of the proposed molten pool image-based monitoring system.

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Quality assurance

System modules cont.

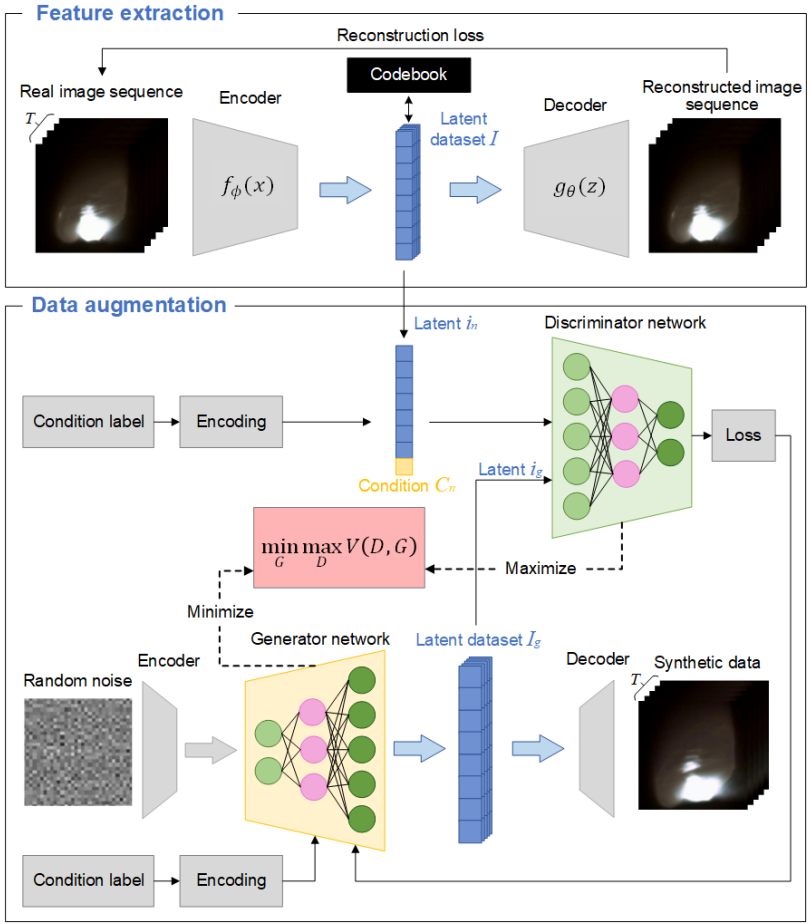
- Data enhancement

Algorithm

- Vector quantized variational autoencoder (VQVAE)
- conditional generative adversarial networks (CGANs)

$$\min_G \max_D V(D,G) =$$
$$\mathbb{E}_{x\sim p_{data}(x)}[\log D\,(x|y)]$$
$$+\,\mathbb{E}_{z\sim p_z(z)}\left[\log\left(1-D\left(G(z|y)|y\right)\right)\right]$$

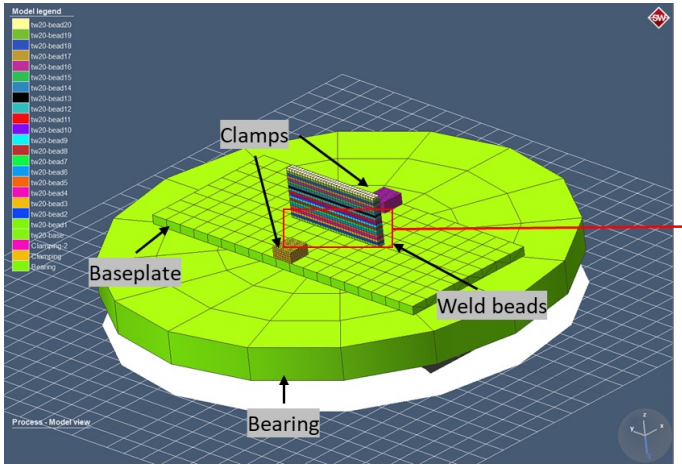
z: random noise; y: condition

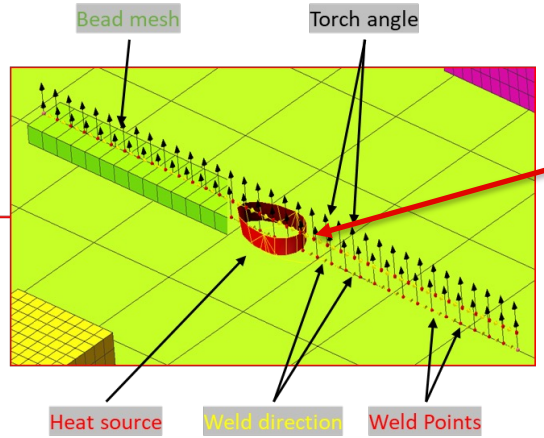


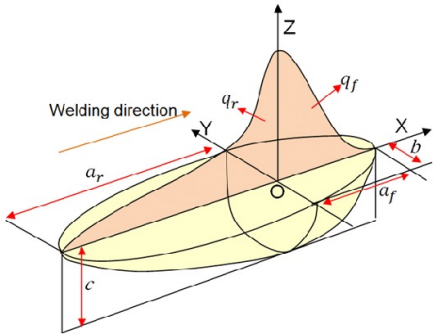
VQVAE feature extraction and CGAN data augmentation modules.

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Numerical simulation: FEM model





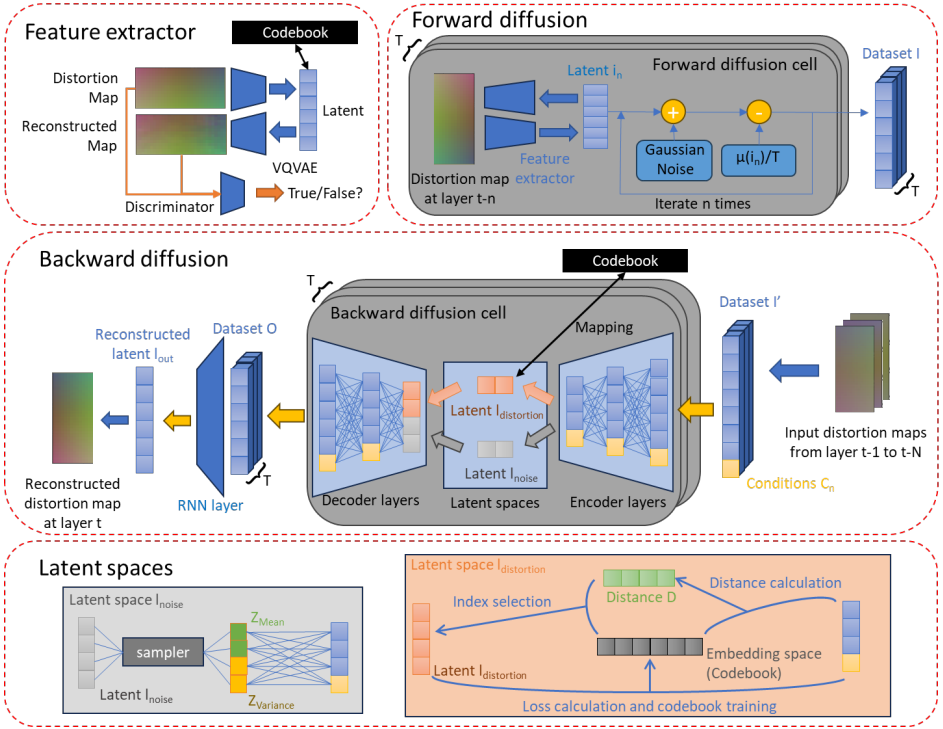


Goldak's heat source model

FEM simulation setup in Simufact Welding

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ML-based surrogate model for fast simulation



Feature extractor: a Vector Quantized Variational Autoencoder Generative Adversarial Network (VQVAE-GAN) is trained to represent the 2D distortion map via a 1D array for dimension reduction.

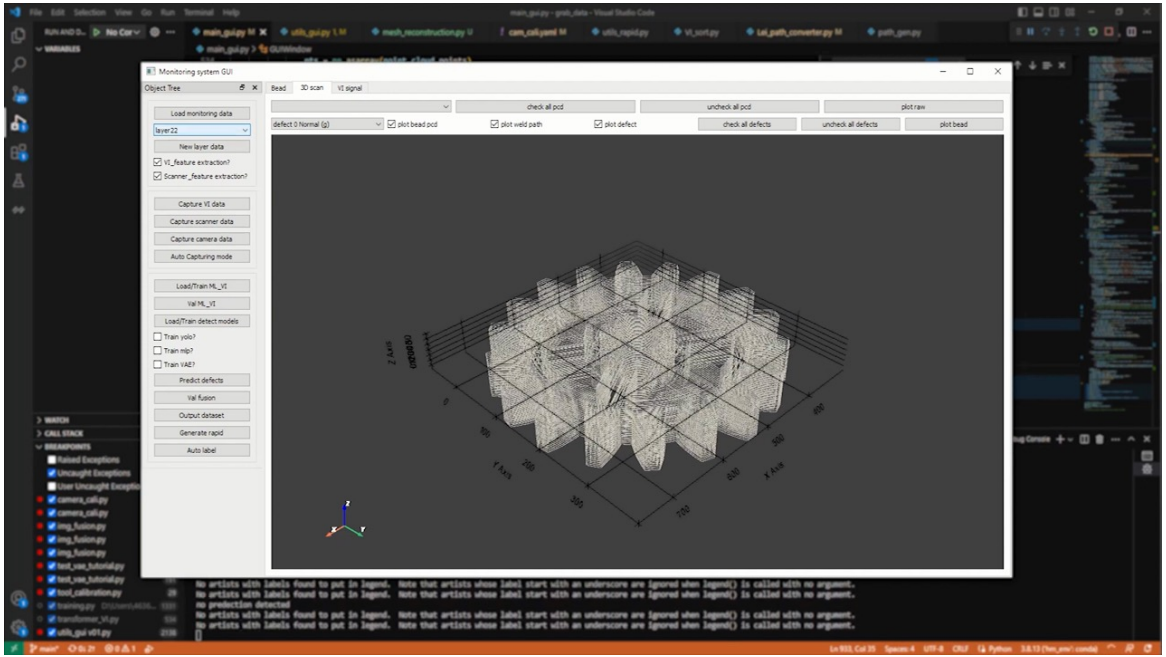
Forward diffusion: simulate the distortion iteration in a layer-by-layer manner. Noise is added to blur the distortion of most previous (bottom) layer as it has smallest effect to distortion changes of current (top) layer.

Backward diffusion: aims to reverse the forward diffusion process, where the output is predicted by distortion history and deposition information of current and previous layers**.**

Latent space design: A combination of VQVAE and VAE where VQVAE aims to model the nonlinear iteration of distortion discretely while VAE aims to denoise (noises are added in the forward diffusion process).

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Fabrication of a Large Gear



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